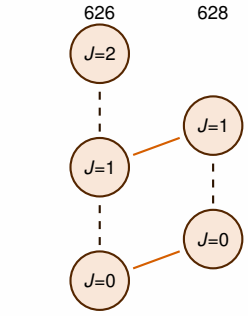
1. Physical Graph Construction



Node = energy state
Features: E, J, parity, etc.

— inter-isotopologue (perturbation)

- - intra-isotopologue (■J=1 ladder)

2. GraphSAGE Architecture

Input graph

↓

GNN + DNN

↓

Softmax → $\hat{p}_c$ per class

3. Hungarian Uniqueness Solver

Cost $C_{ij} = 1 - P_{ij}$,
per (isotope, J, parity, P) block

	c_1	c_2	c_3
s_1	0.10	0.70	0.85
s_2	0.15	0.40	0.90
s_3	0.60	0.55	0.20